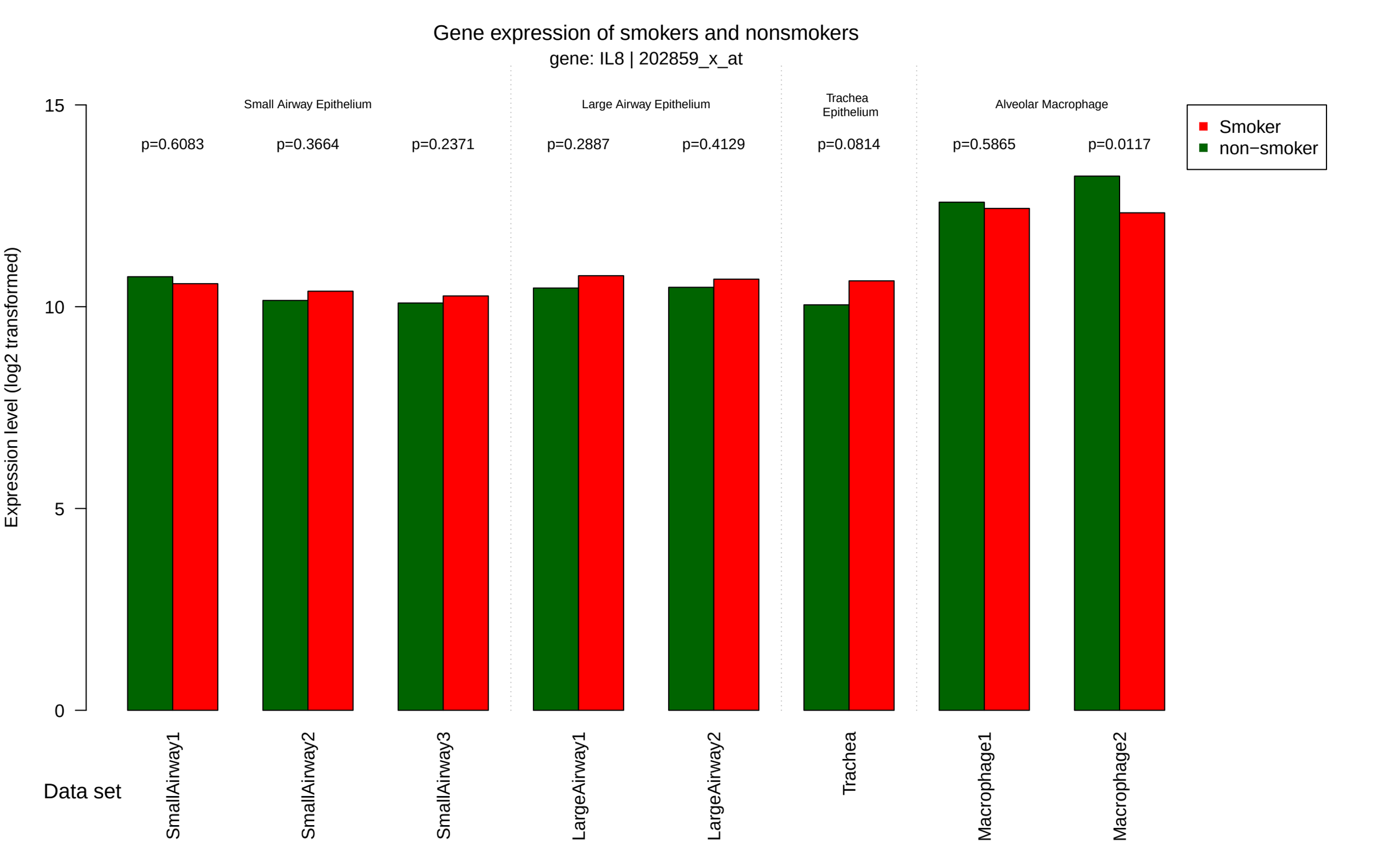
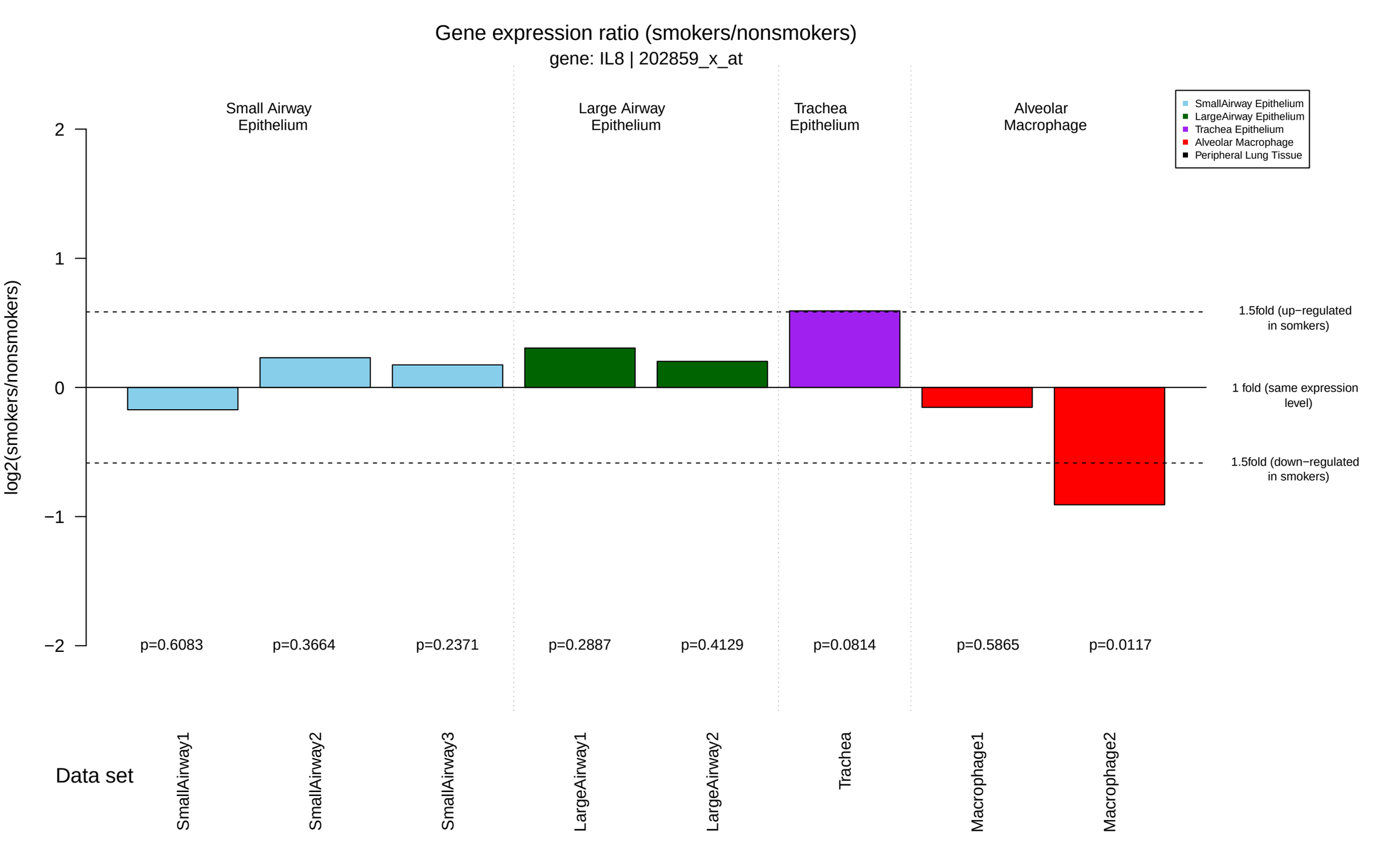
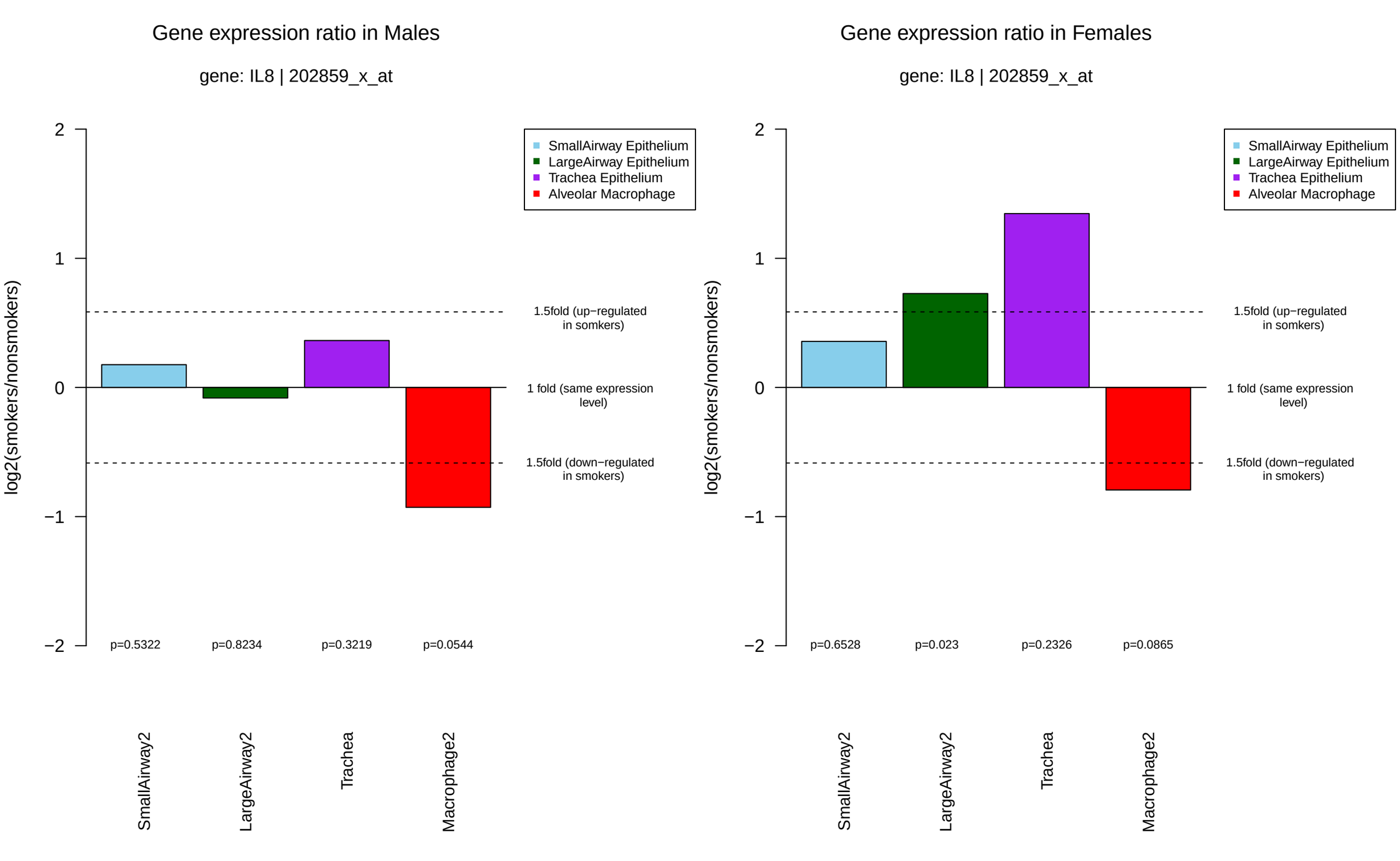
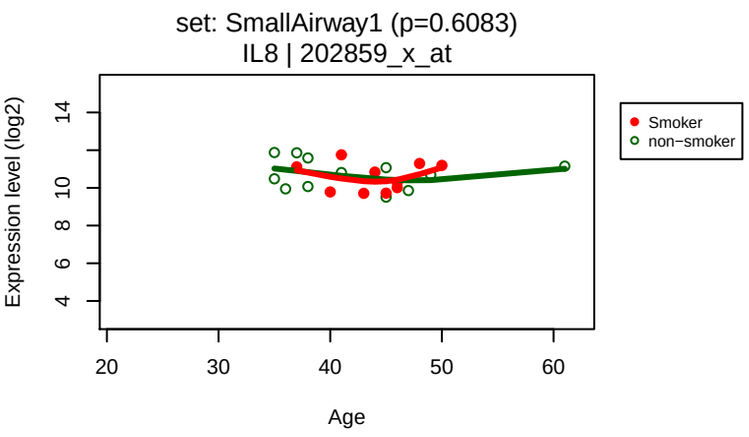
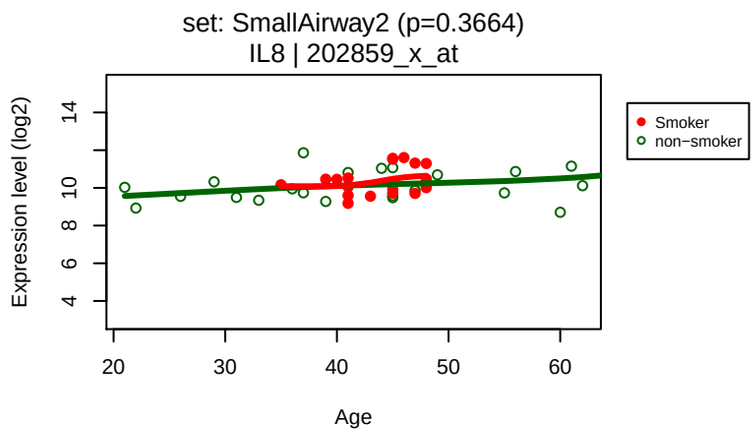
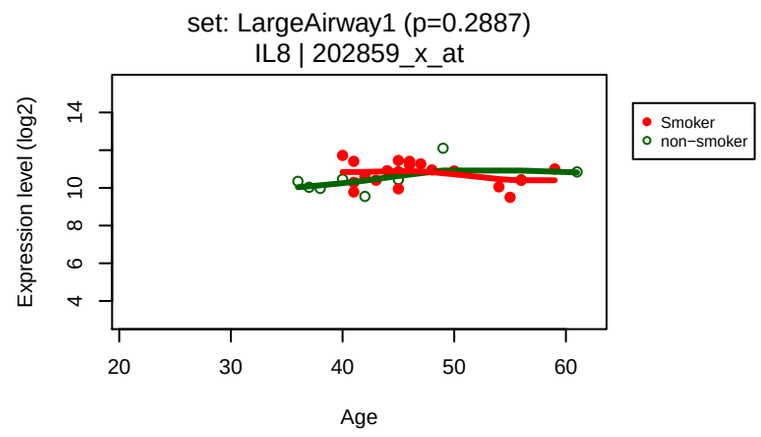

Gene expression of smokers and nonsmokers
gene: IL8 | 202859_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6083
p=0.3664
p=0.2371
p=0.2887
p=0.4129
p=0.0814
p=0.5865
p=0.0117
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

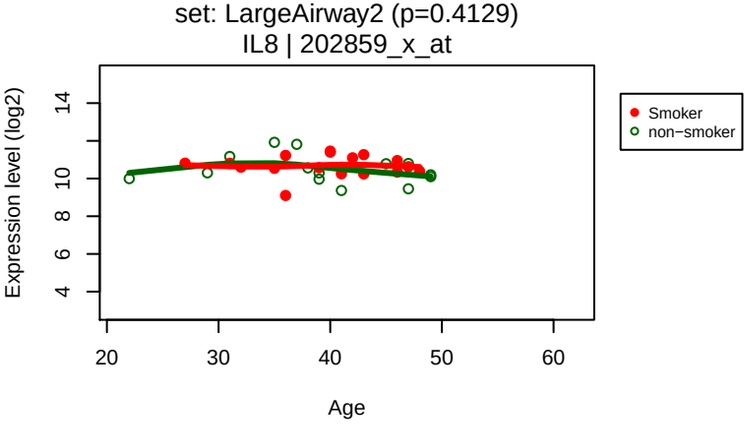

Gene expression ratio (smokers/nonsmokers)
gene: IL8 | 202859_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6083
p=0.3664
p=0.2371
p=0.2887
p=0.4129
p=0.0814
p=0.5865
p=0.0117
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

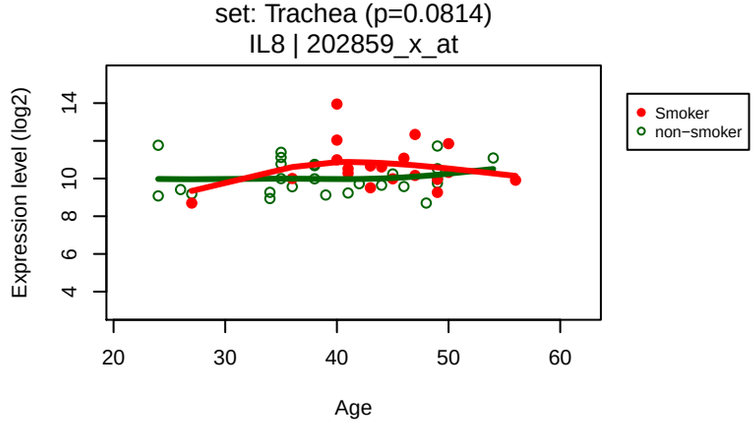

Gene expression ratio in Males
gene: IL8 | 202859_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5322
p=0.8234
p=0.3219
p=0.0544
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL8 | 202859_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6528
p=0.023
p=0.2326
p=0.0865
SmallAirway2
LargeAirway2
Trachea
Macrophage2

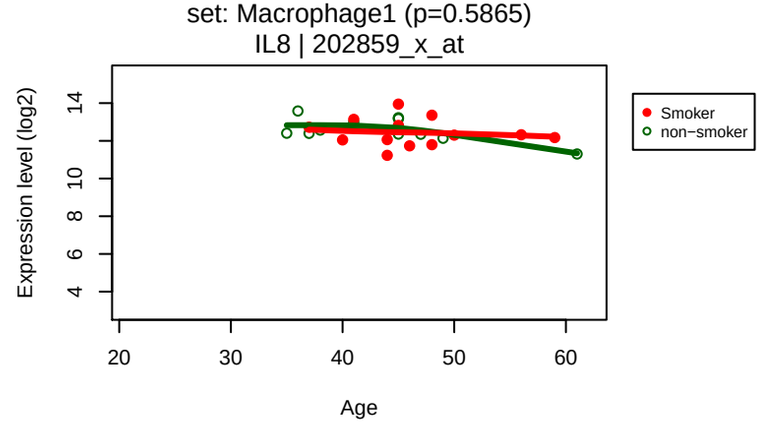

set: SmallAirway1 (p=0.6083)
IL8 | 202859_x_at
Expression level (log2)
Age
Smoker
non−smoker

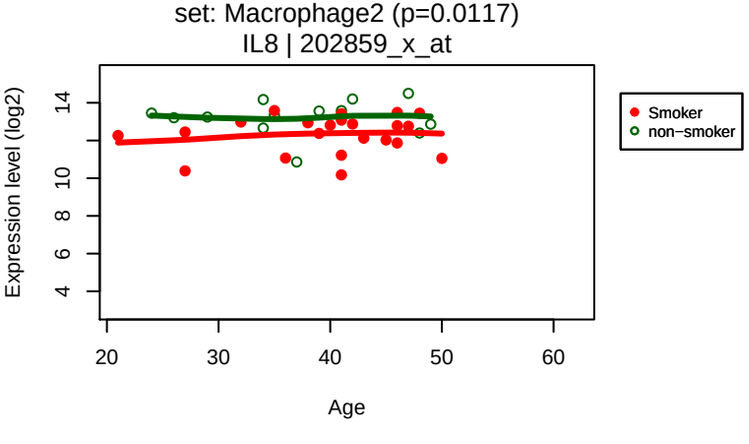

set: SmallAirway2 (p=0.3664)
IL8 | 202859_x_at
Expression level (log2)
Age
Smoker
non−smoker

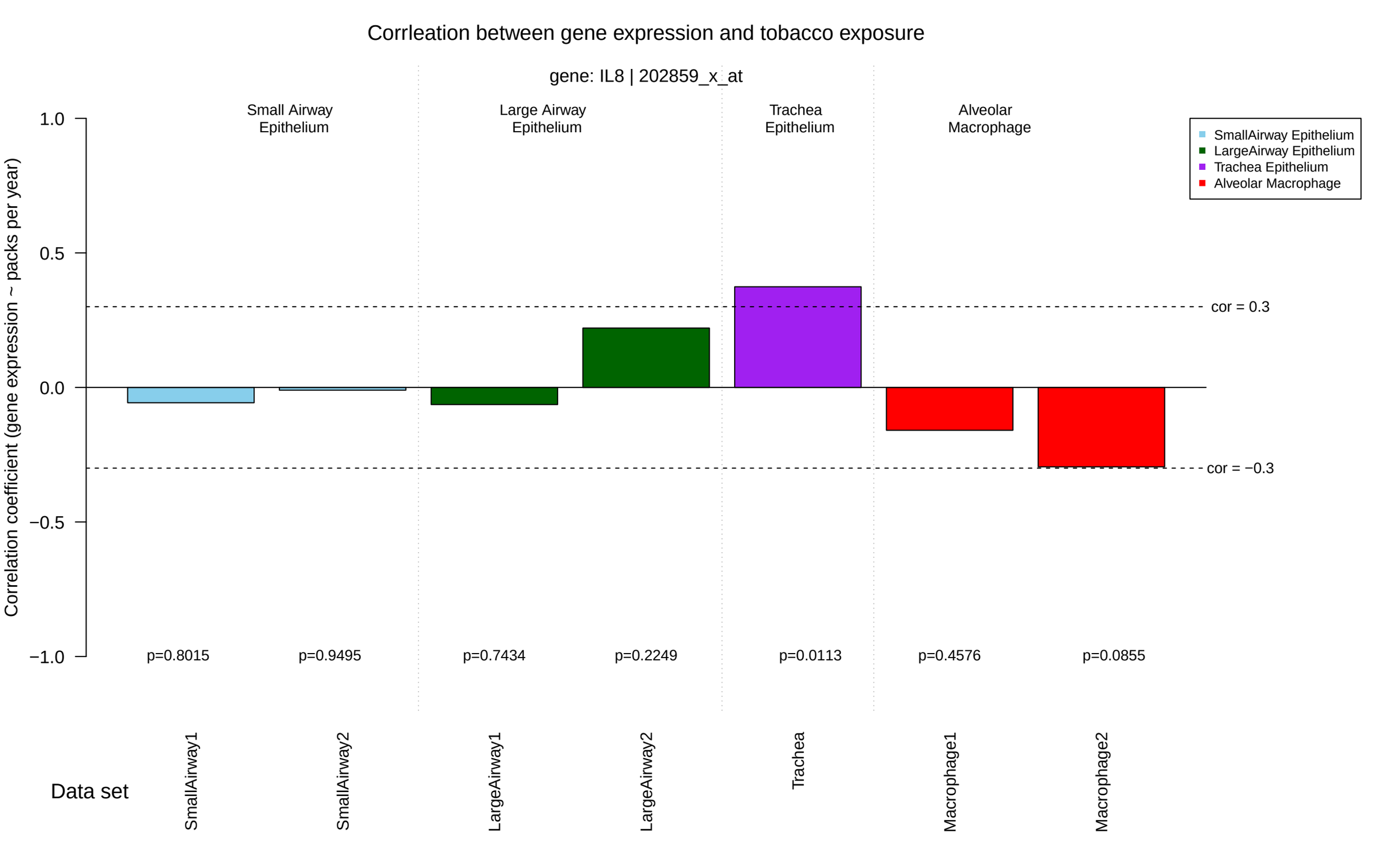

set: LargeAirway1 (p=0.2887)
IL8 | 202859_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4129)
IL8 | 202859_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0814)
IL8 | 202859_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5865)
IL8 | 202859_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0117)
IL8 | 202859_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL8 | 202859_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8015
p=0.9495
p=0.7434
p=0.2249
p=0.0113
p=0.4576
p=0.0855
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2